

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

May 2014

CL 406 Design of Structures - II

Date: 07.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, IS 875:1987, SP -16, Steel Table are allowed.

SECTION - I

- Q - 1 (a) Design a continuous reinforced concrete beam of rectangular section to support dead load of 10 kN/m and live load of 12 kN/m over 3 spans of 5 m each. The ends are simply supported. Take M 25 grade concrete and Fe 415 steel. Sketch the details of reinforcements in the beam. [07]
- Q - 2 (a) Design a trapezoidal footing for the two columns A and B transmitting service loads of 700 kN and 1400 kN respectively. The column A is 300 mm square and column B is 500 mm square in size and they are spaced at 4.5 m centers. The property line is 300 mm beyond the face of column A. The safe bearing capacity of soil at site is 160 kN/m². Use M 25 grade concrete and Fe 415 steel. [07]
- (b) A cantilever retaining wall is used to retain the earth 4 m above ground level. The density of soil is 16 kN/m³ and its angle of repose is 30°. The embankment is horizontal at top. The safe bearing capacity of soil is 180 kN/m² and the coefficient of friction between soil and concrete is 0.5. Adopt M 20 grade concrete and Fe 415 HYSD bars. Design the stem and do necessary checks. [07]

OR

- Q - 2 (a) Design a rectangular combined footing connecting two columns A and B, 4.5 m centre to centre, carrying an ultimate axial load of 1200 kN and 1600 kN respectively. The boundary line of the property is 450 mm from the outer face of the column A. Column A is 380 mm × 380 mm and column B is 440 mm × 440 mm. The bearing capacity of the soil obtained from plate load test is 100 kN/m². Use M 25 grade concrete and Fe 415 steel. Assume effective cover equal to 60 mm. [07]
- (b) Design a counterfort retaining wall based on the following data: [07]
- Height of wall above ground level = 5.5 m
- SBC of soil = 150 kN/m²

Angle of internal friction $\phi = 30^\circ$

Density of soil = 15 kN/m^3

Spacing of counterfort = 3 m c/c

Adopt M 25 grade concrete and Fe 415 HYSD bars

Design the counterfort and do necessary checks.

- Q - 3 (a) Design the longitudinal steel for a braced column $550 \times 450 \text{ mm}$ bent in single curvature with the following data: [07]

Effective length about both axes = 6.0 m , unsupported length = 7.0 m , factored load = 2800 kN , factored moments on major axis are 280 kNm at the top and 220 kNm at bottom. The factored moments about minor axis are 140 kNm at top and 120 kNm at the bottom. Take M 35 grade concrete and Fe 415 steel.

- (b) A longitudinal type of staircase spans a distance of 4 m centre-to-centre of beams. [07]
The rise $R = 150 \text{ mm}$, going $G = 250 \text{ mm}$ and tread $T = 275 \text{ mm}$. The treads have 15 mm granolithic finish and consist of 15 steps. Assuming grade of concrete as M 25 and grade of steel as Fe 415, design the staircase for a live load of 4.5 kN/m^2 . Assume breadth of staircase as 1.6 m .

OR

- Q - 3 (a) Design a RCC column of size $400 \text{ mm} \times 600 \text{ mm}$ for a factored axial load of 2400 kN . [07]
The unsupported length of the column is 6.5 m . For bending about major axis, the column is fixed at the base and hinged at the top. For bending about the minor axis, the column is hinged at both ends. Take M 25 grade concrete and Fe 415 steel.

- (b) A staircase slab spanning longitudinally is set into pockets of two supporting beams, [07]
one of either end, in successive landings between the floors. If the effective span of the slab is 3.5 m horizontally and the rise of the stair is 1.75 m with 250 mm treads and 150 mm risers, design and detail the staircase slab. Assume a live load of 4 kN/m^2 , finishes 0.5 kN/m^2 , $f_{ck} = 25 \text{ MPa}$ and $f_y = 415 \text{ MPa}$.

SECTION – II

- Q - 4 (a) A column ISHB 400 @ 759.3 N/m carries an axial factored compressive load of 2000 kN [07]
and factored bending moment of 100 kN-m . Design the base plate only. Assume concrete pedestal of M20 grade.

- Q - 5 (a) A single storey workshop building situated in a partially developed industrial area is [05]
 $16 \text{ m} \times 12 \text{ m}$ to be provided with a pitch roof. The c/c spacing of truss is 4 m . The sloping roof has a span of 12 m and pitch $1/4$. The height of eaves is 5 m above

ground level. The building will be situated in Bombay in a plane area and its permeability is normal. Determine the wind pressure.

- (b) A gantry girder is to be used in an industrial building for the following data: [09]

Crane capacity – 100 kN

Weight of the crane girder excluding crab – 160 kN

Weight of crab – 35 kN

Minimum clearance between crane hook and the gantry girder – 1.0 m

Wheel base – 3.0 m

c/c distance between gantry rails – 20 m

c/c distance between columns – 6 m

Self-weight of rail section – 300 N/m

Diameter of crane wheels – 150 mm

Steel is of grade Fe 410.

Determine the maximum shear force and bending moment for moving loads. Also give the trial section for gantry girder.

OR

- (b) Determine the maximum bending moment and shear force for the moving loads. The position of wheel load is shown in figure 1 for a gantry girder. Also give the trial section for gantry girder and gross moment of inertia for the built-up section. [09]

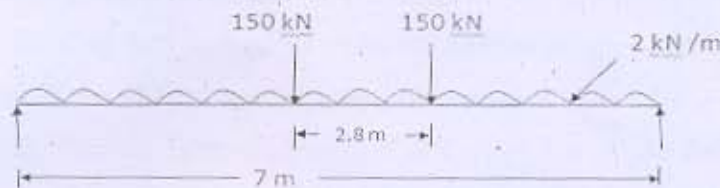


Figure: 1

- Q - 6 (a) A sloping roof truss has the following dimensions: [05]

Spacing of truss = 4 m, Spacing of purlins = 1.4 m, Angle of roof truss = 26.5°

Dead Load = 0.5 kN/m

Live Load = 0.6 kN/m

Wind Load (Suction) = 1.2 kN/m

Use Channel section as a purlin. Do the check for moment capacity.

- (b) A welded plate girder for an effective span of 30 m and carrying a uniformly distributed load of 120 kN/m throughout the span exclusive of its self weight. The girder is laterally restrained throughout. Design the cross section and do the checks [09]

for bending strength and shear capacity of web. The steel for the flanges and web plates is of grade Fe 410.

OR

- (b) A welded plate girder of span 18 m has following elements:

[09]

Flange plates 400 x 16 mm, one plate for each flange

Web 2000 x 10 mm.

Compute the sectional properties and moment of resistance of the plate girder. Design also the load carrying and end bearing stiffeners, if the plate girder is to carry uniformly distributed load of 120 kN/m. Steel is of grade Fe 410.
